

ORDER OUTCOME ANALYSIS AND TRADE MARKING IN HIGH-FREQUENCY TRADING

1. MOTIVATION

We use the term order outcome analysis (OOA) to refer to analyzing cases where our order volume is sufficiently large to admit statistical analysis, usually in the case of mid- to high-frequency trading. OOA is used by market participants trading on their own account, and also published (or, some statistical metrics) are published to brokerage clients relying on broker execution. In the latter case, the analysis is typically presented only on trades, rather than all orders placed, and we use the term *post-trade analysis* or PTA.

Analyzing order action is a classic example of split-apply-combine analysis.

2. TRADE MARKING

When we place orders, some of them will result in partial or total *fills*, i.e. trades. For these, we receive an *execution report* from the exchange with order id number, price, quantity, and usually other fields such as exchange timestamp.

To begin, we decorate our trades with data that is clearly important, or might be important. Some of this must be done live, at the time of the trade (or by employing very extensive systems logging), other items can be added later, and *markouts* must be added later.

We will use $\mathcal{K} = \{k_1, k_2, \dots, k_N\}$ denote the set of all trades executed over an analysis period.

2.1. Definition. We can think in terms of *marking functions* $m : \mathcal{K} \rightarrow \mathcal{M}$ that assign metrics to each trade k_i . Some marks are simple, such as whether we placed our traded order with aggressive or passive intent (or with some exchange-defined combination of the two). Other definitions, especially categorical ones, are a matter of taste such as whether a trade occurred during a market activity burst or not.

It is often convenient to avoid categorical assignments, especially ones derived from market state, at trade marking time, saving that task for later analysis.

Note that in a limit order book, an aggressive order may interact with *multiple* passive orders, generating a separate trade report for each of them. In this case the official exchange timestamps should be identical, but the information may come in multiple network packets, yielding distinct local timestamps.

In some situations, especially US equity trading, short sale rules can also cause a trade report to be split in two.

In multiple-hit cases, it frequently makes sense to form single marks by summing all the individual quantities, taking a quantity-weighted average as the price, and using the first sub-trade timestamps.

2.2. Typical Marks. Effective order marking systems often employ multiple orthogonal dimensions:

2.2.1. *Order Type Marking.* Classifies trades by how they interacted with the order book:

$$m_{\text{type}}(t_i) = \begin{cases} \text{P or PASSIVE} & \text{if } t_i \text{ provided liquidity} \\ \text{A or AGGRESSIVE} & \text{if } t_i \text{ removed liquidity} \end{cases}$$

In practice, exchanges offer a wide variety of order flags, including P into A and A into P, as well as flags allowing midpoint, iceberg, and hidden for passive orders. Good systems will record all these flags for all orders.

2.2.2. *Signal Strength Marking.* For signal-driven strategies, mark orders by the magnitude of the predictive signal at execution time. If $\vec{\alpha}(\tau_i)$ is a vector of various signal values

$$m_{\text{signal}}(t_i) = \vec{\alpha}(\tau_i)$$

We often partition these into quintiles or other quantiles during post-processing. If we do so without lookahead, we can have direct means of improving associated trading.

2.2.3. *Order Signal Triggers.* As of the last *decision time* preceding a trade report, the system's most recent update was typically prompted by some sort of signal. We can record the market instrument that prompted it, the exchange, or any other characteristics of that prompt.

This is often a feature helping reveal *rares*, as well as positions in the liquidity ladder.

2.2.4. *Market State.* We can take a simple metric of volatility

$$m_{\text{vola}}(t_i) = \text{EMS}_{\lambda} \left(\left(\frac{p_t - p_{t-1}}{p_{t-1}} \right)^2 \right)$$

or a recent count of exchange messages J_t

$$m_{\text{burst}}(t_i) = \#\{J_t \ni t_i - \Delta T < t < t_i\}$$

Again we may later turn these into categorical quantiles

2.2.5. *Spread Marking.* Relative bid-ask spread is very important:

$$m_{\text{spread}}(t_i) = \frac{p^{\text{ask}}(\tau_i) - p^{\text{bid}}(\tau_i)}{p^{\text{mid}}(\tau_i)}$$

We can also include metrics of book depth at quantity.

3. PL PERFORMANCE

3.1. **P&L Attribution.** For each mark m , usually restricted to trades only, compute the aggregate P&L marked to set of horizons $\mathcal{H} = \{h_1, \dots\}$. This depends on a choice of future mark price. Pessimistically this would be the bid for long trades and offer for short trades. More commonly, a mid price or micro price is considered best.

$$\text{P\&L}^h(t_i) = q_i \cdot (p_i^h - p_i) - c_i$$

c_i represents transaction costs (fees, rebates).

If we choose an inventory system, such as FIFO or LIFO, we can also choose exit as a horizon and use p_i^{exit} , the exit price, gives us a solid mark, though to a clock time horizon inconsistent among trades. The marked P&L is:

$$\text{P\&L}_m = \sum_{t_i \in \mathcal{T}_m} \text{P\&L}(t_i)$$

3.2. Win Rate and Profit Factor. Win rate for mark m :

$$W_m = \frac{|\{t_i \in \mathcal{T}_m : \text{P\&L}(t_i) > 0\}|}{|\mathcal{T}_m|}$$

Profit factor (ratio of gross profits to gross losses):

$$\text{PF}_m = \frac{\sum_{t_i \in \mathcal{T}_m^+} \text{P\&L}(t_i)}{-\sum_{t_i \in \mathcal{T}_m^-} \text{P\&L}(t_i)}$$

where $\mathcal{T}_m^+ = \{t_i \in \mathcal{T}_m : \text{P\&L}(t_i) > 0\}$ and $\mathcal{T}_m^- = \{t_i \in \mathcal{T}_m : \text{P\&L}(t_i) < 0\}$.

3.3. Slippage Analysis. We say *slippage* is the difference between execution price and decision price. For trade t_i with decision price p_i^{decision} :

$$\text{Slip}^h(t_i) = \text{sign}(q_i) \cdot (p_i^h - p_i^{\text{decision}})$$

4. PLOTTING

Our marks can be used in feedback loops for improving algorithmic trading entirely without human intervention. However, the most important target of this work is human understanding and intervention.

4.1. P&L Term Structure. Our most basic plots aggregate P&L at the various horizons. We can make plots with horizon on the horizontal axis, and average PL and/or quantiles of PL on the vertical one.

Ultimately, we have empirical probability densities at all horizons, so we can analyze other statistics such as tail metrics.

4.2. Category Splitting. Many of our decorations are already categorical (P/A) or can easily be made so (vola quintiles). We frequently construct multiline plots splitting results over these categories.

Such plots can suggest unprofitable orders to avoid, or highly profitable ones to exploit.